Evolution of Pregnancy Detection from Prehistoric Era to Modern Era

Pregnancy detection has evolved significantly from ancient times to the modern era. Early methods relied on physical signs, natural substances, and cultural beliefs, often with low accuracy. With the advancement of medical science, experimental techniques and urine analysis emerged.
The discovery of the hormone hCG revolutionized pregnancy testing, leading to modern immunoassay-based methods that are fast, reliable, and accessible. This presentation explores the historical evolution and scientific progress of pregnancy detection.

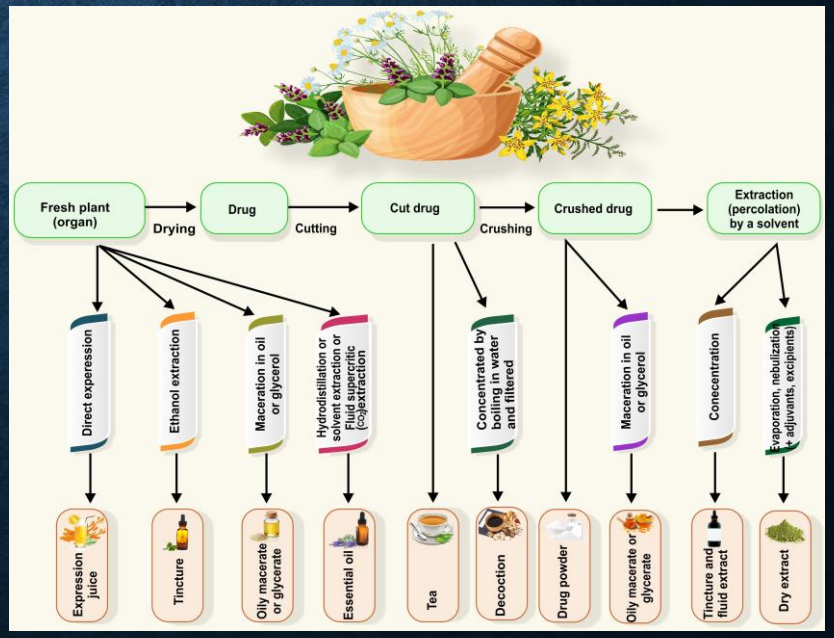
PREHISTORIC ERA – PREGNANCY DETECTION

Onion Test

•An onion was placed in the vagina overnight.
•The woman's breath was checked the next morning. Onion smell in the mouth meant not pregnant.
•No smell was believed to indicate pregnancy.

Wheat & Barley Test

•A woman urinated on wheat and barley seeds.
•Seeds were observed over several days. Sprouting of seeds indicated pregnancy.
•No sprouting suggested not pregnant.



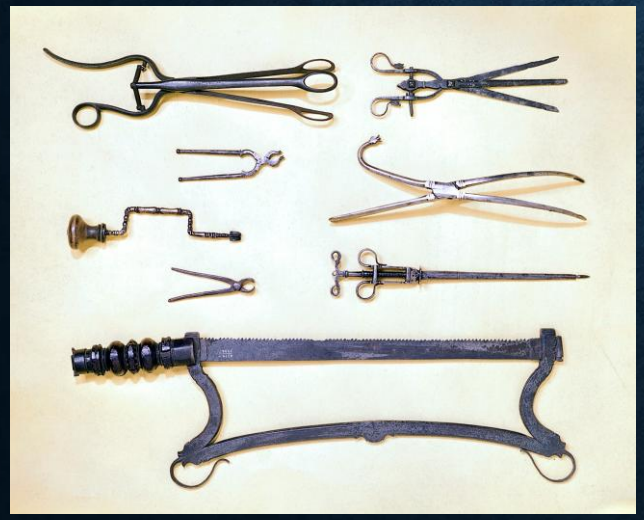
MIDDLE ERA – UROSCOPY

Matrons or Nurses, *circa* 1300. Willemin, 'Mon. inédits.'

UROSCOPY

- A woman's urine was collected in a transparent glass flask called a matula.
- The urine was examined visually under light. Doctors observed the color, clarity, thickness, foam, and sediment of the urine.
- Certain urine appearances, such as pale, cloudy, or milky urine, were believed to indicate pregnancy.
- No chemical reactions or measuring instruments were involved.
- Diagnosis depended on experience, observation, and medical beliefs of the time.

RENAISSANCE ERA PREGNANCY TEST



Wine / Vinegar Method

•The test was used during the Renaissance period as an early method to detect pregnancy.
•A woman's urine was collected in a container.
•Wine or vinegar was added to the urine.
•The mixture was left undisturbed for some time.
•If the liquid became cloudy or milky, pregnancy was assumed.
•The cloudiness occurred due to reaction between alcohol and increased proteins in pregnant urine.

This method reflected early experimental medical thinking but was unreliable and later replaced by hormone-based tests.

MODERN ERA PREGNANCY DETECTION

Urine hCG Test - Biomedical Approach

- Used in the modern era as a reliable and non-invasive method for pregnancy detection.
- A woman's urine is collected, and a test strip or cassette is dipped into the sample.
- The test detects hCG (human chorionic gonadotropin), a biological biomarker released after embryo implantation.
- The strip contains monoclonal antibodies that specifically bind to hCG (immunoassay-based biosensor).
- Antigen-antibody binding triggers a biochemical reaction. This reaction is converted into a visual signal (two lines or positive symbol).
- The process represents signal transduction from a biological input to a readable output.

CONCLUSION

The journey of pregnancy detection highlights the growth of human curiosity, medical knowledge, and technological innovation. From ancient observational practices and ritualistic tests to laboratory experiments and hormone-based diagnostics, each era contributed to improving accuracy and understanding. The discovery of hCG and the development of immunoassay techniques revolutionized pregnancy testing, making it non-invasive, rapid, and highly dependable. Modern home pregnancy kits now allow early detection with convenience and privacy.

Overall, the evolution of pregnancy detection reflects not only scientific advancement but also progress in women's healthcare. As technology continues to develop, future methods may become even earlier, smarter, and more precise.